1 Some background

in a nutshell...

I (Josh) am in the process of completing my MSc, which has been about trying to search for *slow radio transients* with an important upcoming Canadian radio telescope called *CHORD*. CHORD is a very sensitive instrument with a huge field of view, which is great for detecting these signals, but it has kind-of 'blurry' vision, so the transient signals we want to find get mixed in with all of the uninteresting other radio sources in the sky. The telescope sees them all as overlapping, and often can't tell if something was a real transient signal, or just a background source that got a little brighter. This is a textbook problem for some machine learning algorithms, which we want to try! We've simulated what the sky will look like to CHORD, and now we want to try using ML on it to find the signals.

a blurb on the science - what are we after?

'Slow radio transient' sounds uninteresting, but this couldn't be farther from the truth; we only call them this so they aren't confused with the famous 'fast radio bursts' (FRBs). The two main sources we're looking for are *Gamma-Ray Bursts* (GRBs) and *Tidal Disruption Events* (TDEs). GRBs come from the deaths of the most massive stars. The core of the star collapses into a black hole, and the collapsing remainder of the star is accreted very quickly. In a TDE, a star is shredded by tidal forces as it passes an existing supermassive black hole, and so again a ton of material is suddenly accreted. These processes release enormous amounts of energy on a very short timescale; accretion onto a black hole can release 10-40% of the mass energy of the accreted material (compare to $\sim$0.7% for Hydrogen fusion in our sun!). For example, in 2022, GRB 221009A, from 2.6 billion light years away, was so intense that its gamma rays ionized the upper atmosphere, puffing it up so much that it affected spacecraft trajectories.

When a rotating black hole accretes matter this fast, it can launch a relativistic jet. The front of this jet acts like a giant rake, piling up electrons from the surrounding medium which can reach highly relativistic speeds ($\gamma \sim 10 - 1000+!$). These electrons radiate powerfully by a process called *synchrotron radiation*, leading to a burst of radio light that gets brighter as the jet expands, then fades as it loses energy. **This is the 'transient' we want to detect**.

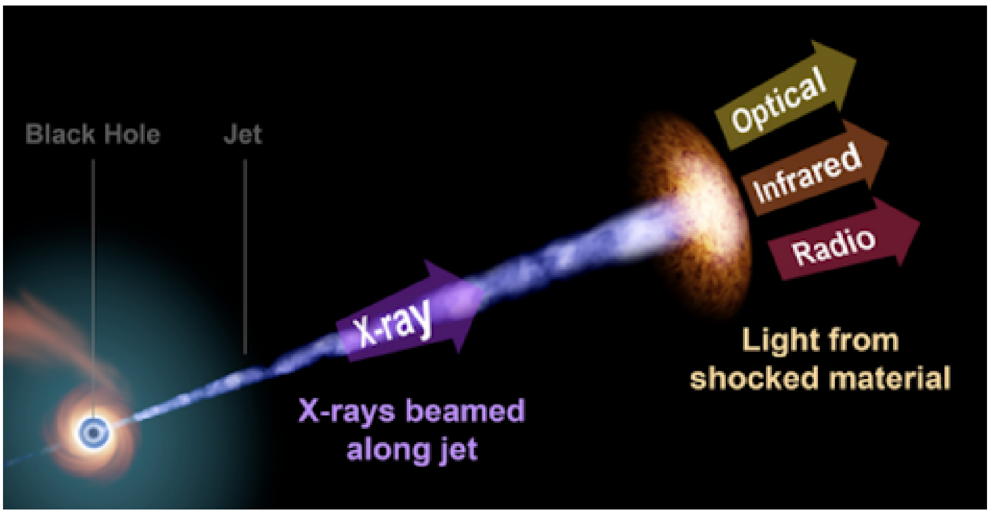

Figure 1: Schematic of the emitting situation with a GRB or TDE. The radio emission is synchrotron emission from the pileup at the front of the jet

Unlike the gamma rays that only last minutes at most, or the optical afterglow that fades over days or weeks, the radio emission from this jet takes months to years to evolve. Radio emission helps us study the jet itself, but it also has another important property. As the jet slows down, radio emission can be observed from *all directions*, whereas the gamma rays and X-rays are tightly focused along the jet axis of the event. This means we can see them from cosmological distances, but if we're not directly in their path, they're invisible at these wavelengths! Dust often heavily obscures optical light from these events as well, leaving radio emission as the only reliable way to find these things if they're not pointed at us.

These kinds of radio transients have been spotted from both gamma ray bursts and tidal disruption events, but they're difficult to find because they're a) very far away and therefore can be quite faint, and b) are very, very rare. There are plenty of radio observatories that can achieve the required sensitivity, and quite quickly, but these have very limited fields of view, so they're usually used to follow up on events that we already spotted with optical telescopes (an example here would be the famous Very Large Array [VLA] in New Mexico). Another class of instrument can see much of the sky at once, but isn't sensitive enough to see the kinds of transient we're after (something like the Allen Telescope Array [ATA] or the Australian Square Kilometre Array Pathfinder [ASKAP], which have a large fields of view, but struggle to reach the necessary sensitivity to detect many of these because they're smaller telescopes, or CHIME, the instrument CHORD is being built next to, which is very sensitive and has an enormous field of view, but has very poor angular resolution, i.e. it sees a very blurry vision of the sky and can't single out the transients).

The challenge is to have a telescope that is both sensitive enough to see the transients, and sees enough of the sky that it will catch them.

Searching with CHORD - pros & cons

Though it wasn't designed for it, CHORD *could* be able to find transients like TDEs and GRBs. On paper, it's certainly sensitive enough, and it has a large field of view. This makes people very interested in the possibility of a search with CHORD. On the other hand, how clearly CHORD sees the sky, its 'angular resolution', is on the edge of being *too* bad for a slow transients search to work. The appearance of the sky at radio frequencies of $\sim$100MHz-10GHz is quite busy; we see whole galaxies, bright active galactic nuclei, supernova remnants and much more. These are all basically persistent sources; they'll be there the whole time during a survey, whereas the transient signal will brighten and fade over months or years. Here's the catch: many of the 'background', non-transient sources also change with time. This isn't a problem if you have a very crisp view of the sky, where you can tell that one source was there the whole time whereas the real transient suddenly showed up where before there was nothing, but if your telescope sees the sky very blurrily, then all of the sources start to stack on-top of one another, and you can't tell whether a brightening-then-fading at some place in the sky was a real transient, or if one or more background sources there all got a little brighter at once. We call this problem of overlapping sources in the sky *source confusion*.

Josh's work

My MSc project has been about evaluating CHORD as a slow-transient detector in the face of these promising features and problems. Besides many pen-and-paper calculations, we've also simulated the transient sources, and CHORD images of them, and the entire rest of the radio sky, as a function of frequency and time. Because we can't individually identify the sources that are this faint, we compute several purpose-built statistics for every spot in the image that are designed to pick out the real signal against the uninteresting background. By doing this we've been able to find many of the real transients, suggesting we could really do this with CHORD, but we also find many 'false positives' where the background conspires to look like a transient, creating a hotspot in the statistics.

We know that these statistics are *an* approach, but they're not provably optimal in any sense. Using other methods, we could find the transients in the simulated images, and maybe eventually the real data, much better!

the gist of your project

The goal of your project will be to investigate how well we can use various machine learning techniques to find the real transients in the images that we've already simulated (and can simulate more of). We'll start simpler than this - as you learn about various machine-learning algorithms, we'll try applying them to situations that are much easier to simulate but are qualitatively similar - a signal we're after which is intermixed with other similar, but not identical behaviour. Of course, it'd be awesome to eventually be able to find more transients than with the non-ML statistical approach, and we should try as hard as we can to achieve that, but ultimately we just want to know whether a machine learning approach would work, and it's possible that it might not be.

resources

machine learning

The most important things for you to understand and know about are various machine-learning algorithms that could work well here. Some good introductions, exposure, and early tools for ML are the following:

- Google Machine Learning Crash Course. Designed to get you up to speed with ML basics, techniques, and key algorithms
- Kaggle. This is a kind of hub for various ML models which offers courses and the like, and also is a place for people to share their models and training datasets
- PyTorch is a big ML library that is largely based on and has a very similar design philosophy to Python.

A great class of algorithms to focus on to start are *convolutional neural networks*. These excel at the kind of problem we have: identifying a particular kind of signal in messy, busy or confusing data (often images!)

Note that I (Josh) am not an ML expert - I will do my best to learn as well, but I will be quite busy with my own projects, especially just as you're starting out, since I will be finishing the writing of my Masters thesis.

radio astronomy and slow transients

What I do know a lot about is radio astronomy, and the kinds of transients we're looking for. If you have questions about this kind of science, I'm always a good person to come to! If you're very interested in this then that's great, but even if not, there will be various concepts you'll need to understand to some degree, since they are parts of the simulated images. If you want to read up more on the science, here are some great resources:

- The CHORD Whitepaper - this covers the basics of what CHORD is, what it is built to do, and how. It isn't as dense as most scientific papers, but in some places it doesn't go into a ton of detail. We'll also want to get you onto the 'CHORD wiki', where collaboration members like Adrian and I store documents, presentations and more with richer details

- NRAO Radio Astronomy Lectures - this is kind of like a 'Cliff Notes' or 'Paul's online math notes' for radio astronomy. It's a little old, but still a great way to get into it. If it suits you better I also have an unrelated textbook I can lend you
- Overview of Modern Radio Astronomy - this review paper covers the state of contemporary radio astronomy. Early sections cover all kinds of sources of radio emission in the sky, and they also cover some of the physics behind it, and various observatories and upcoming/ongoing efforts
- Time-domain Radio Astronomy - this is an excellent and very broad review of 'time domain radio astronomy' - it talks about various kinds of variable and transient sources, and the physics behind what's going on. It's quite long - section 3 is where they describe the most relevant stuff
- Radio Properties of TDEs - this 2020 paper covers, fairly broadly, what we understand about radio observations of TDEs (can be roughly applied to GRBs too). They cover both the observations, and the physics behind what we believe is going on